MODELS, ABSTRACTIONS AND BASIC CONCEPTS

DISTRIBUTED SYSTEMS

Master of Science in Cyber Security

A.Y. 2025/2026



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CYBER INTELLIGENCE AND INFORMATION SECURITY

OUTLINE

- System and Processes;
- Asynchronous Executions;
- Indistinguishable Executions;
- Failures: Crash Stop vs Byzantine;
- Abstractions: Liveness and Safety Properties;
- First Algorithms: Communication Links.
- FIFO LINK

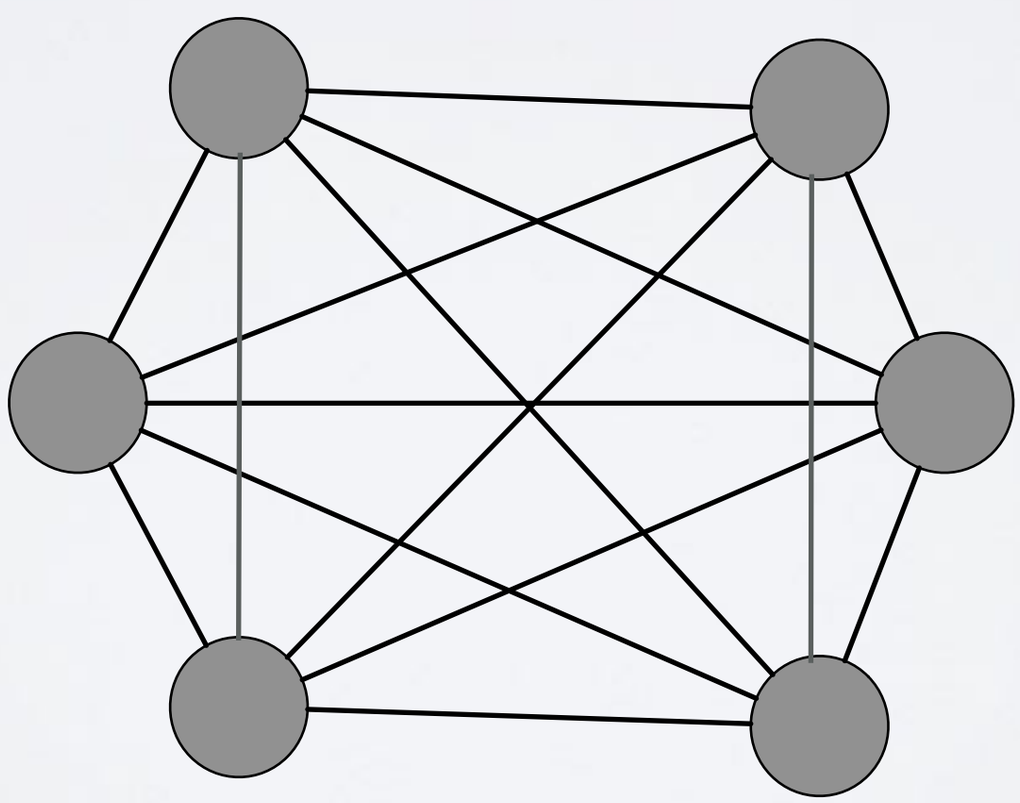


MODEL:

SYSTEM AND PROCESSES

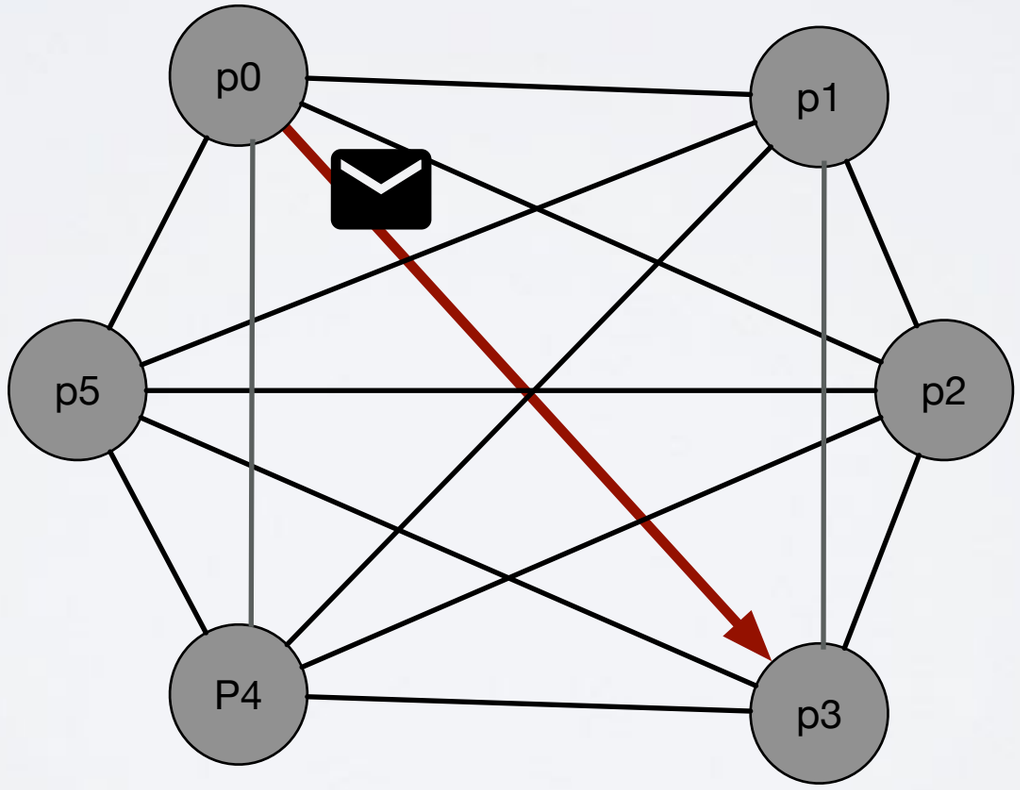
SYSTEM

- We have n processes in $\Pi : \{p_0, \ldots, p_{n-1}\}$ with distinct identities.
- Processes communicate using a communication graph $G : (\Pi, E)$ (usually, G is complete).
- The communication happens by exchanging messages on communication links (edges of G).



SYSTEM

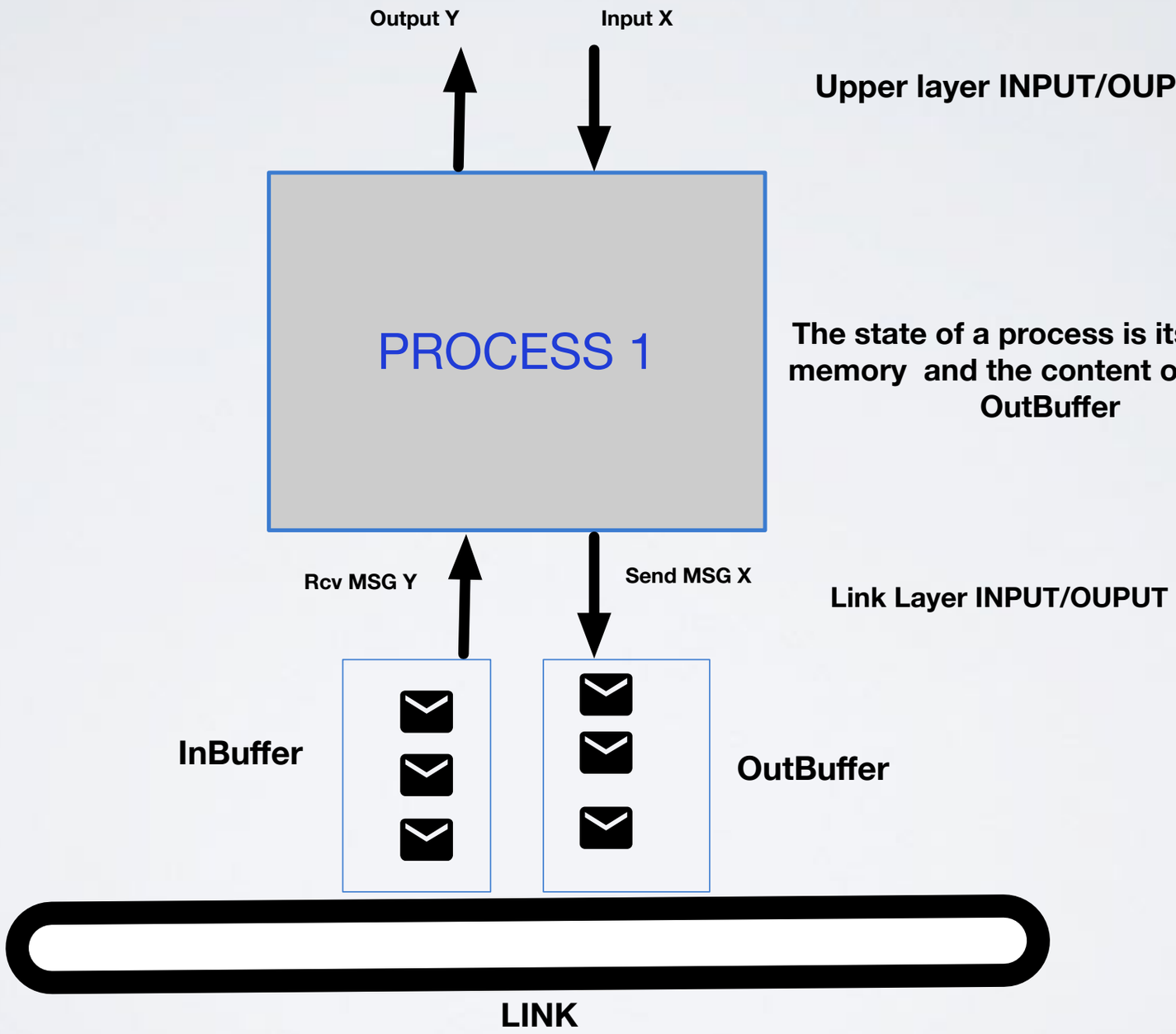
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PROCESS

- A process is a (possibly infinite) State Machine (I/O Automaton).



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PROCESS

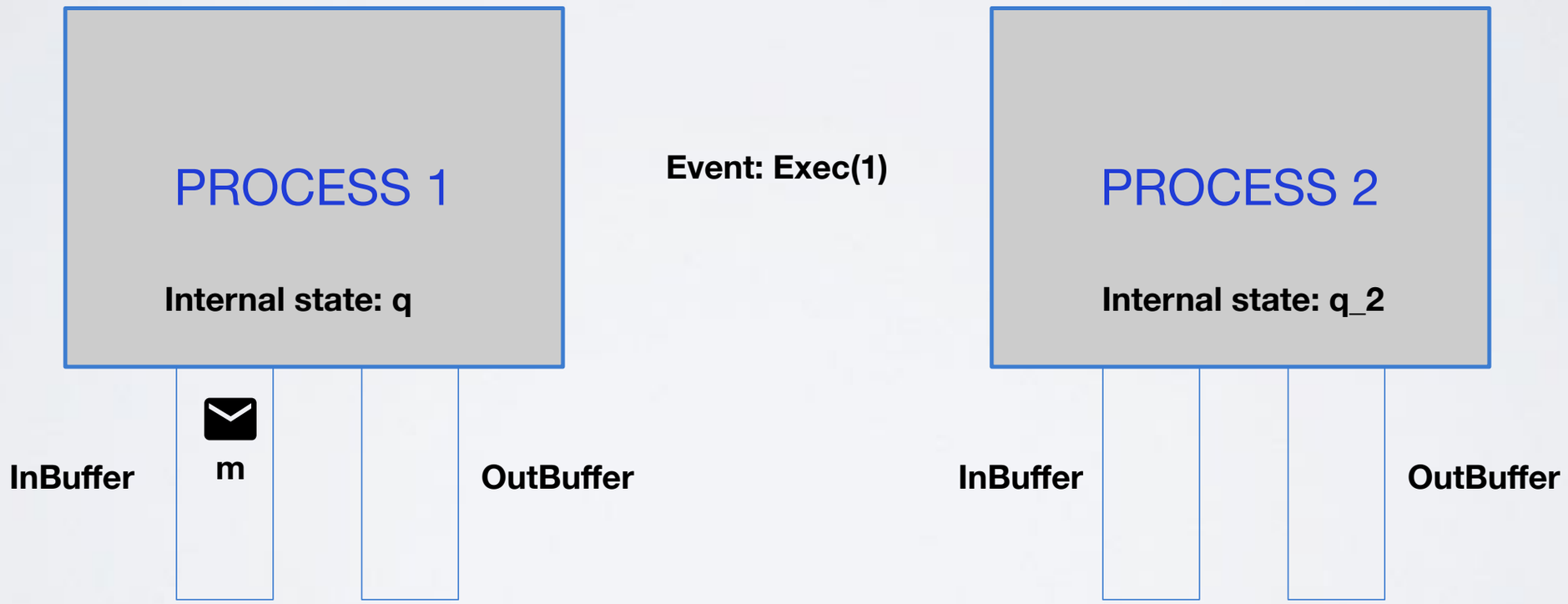
- A process is an infinite state I/O Automaton:
  - Internal states - set Q
  - Initial states - set Qi \subset Q
  - Messages - set of all possible messages M in the form <sender,receiver,payload>
  - InBuf_j - multiset of delivered messages
  - OutBuf_j - multiset of ``inflight'' messages (messages sent but not delivered).

$$P_j(q \in Q \cup Q_{in}, InBuf_j) = (q' \in Q, Send_msg \subset M)$$
$$OutBuf_j = OutBuf_j \cup Send_msg$$
$$InBuf_j = \emptyset$$

MODEL: ASYNCHRONOUS EXECUTIONS

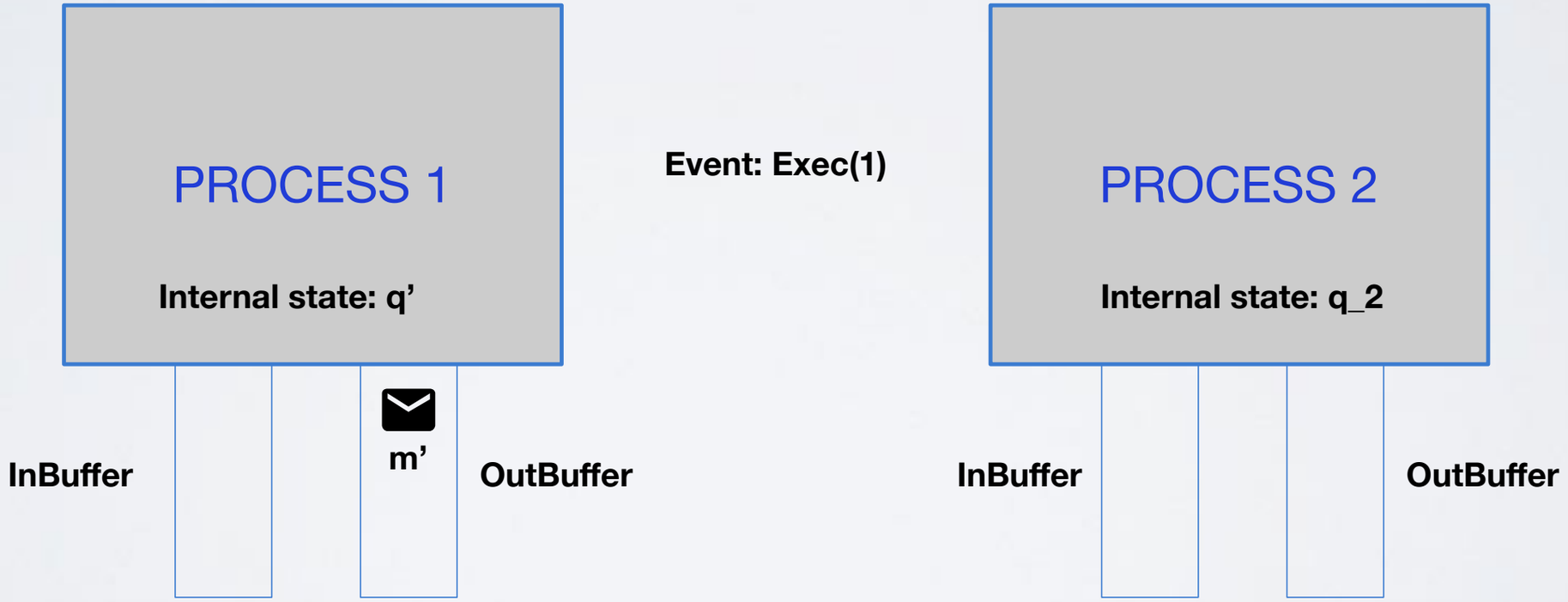
EXECUTION

- There is an adversary that schedule a set of events (scheduler):
  - Delivery of a msg - Del(m,i,j): move message m from OutBuff_i to InBuff_j
  - Execution of a local step - Exec(i): process i executes one step of its state machine.



EXECUTION

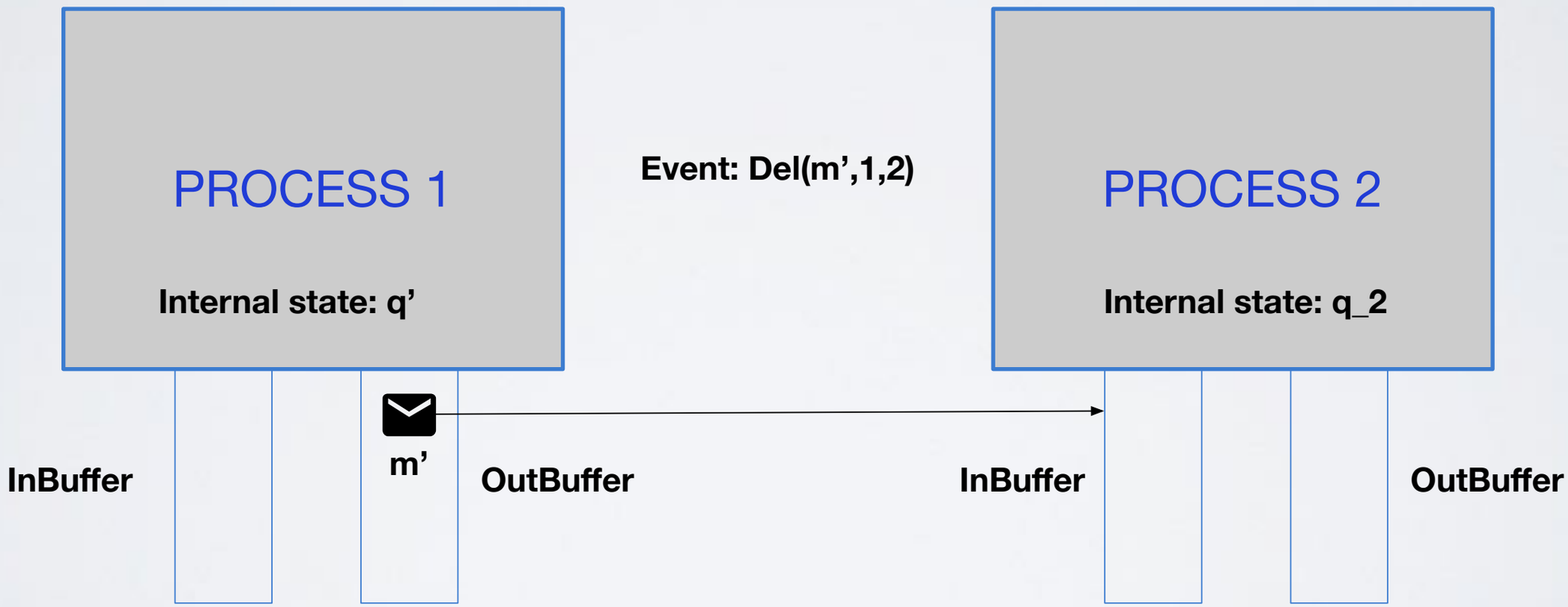
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$$P_1(q, \{m\}) = (q', \{< 1, 2, m' >\})$$
$$OuBuf_1 = OutBuf_1 \cup \{< 1, 2, m' >\}$$

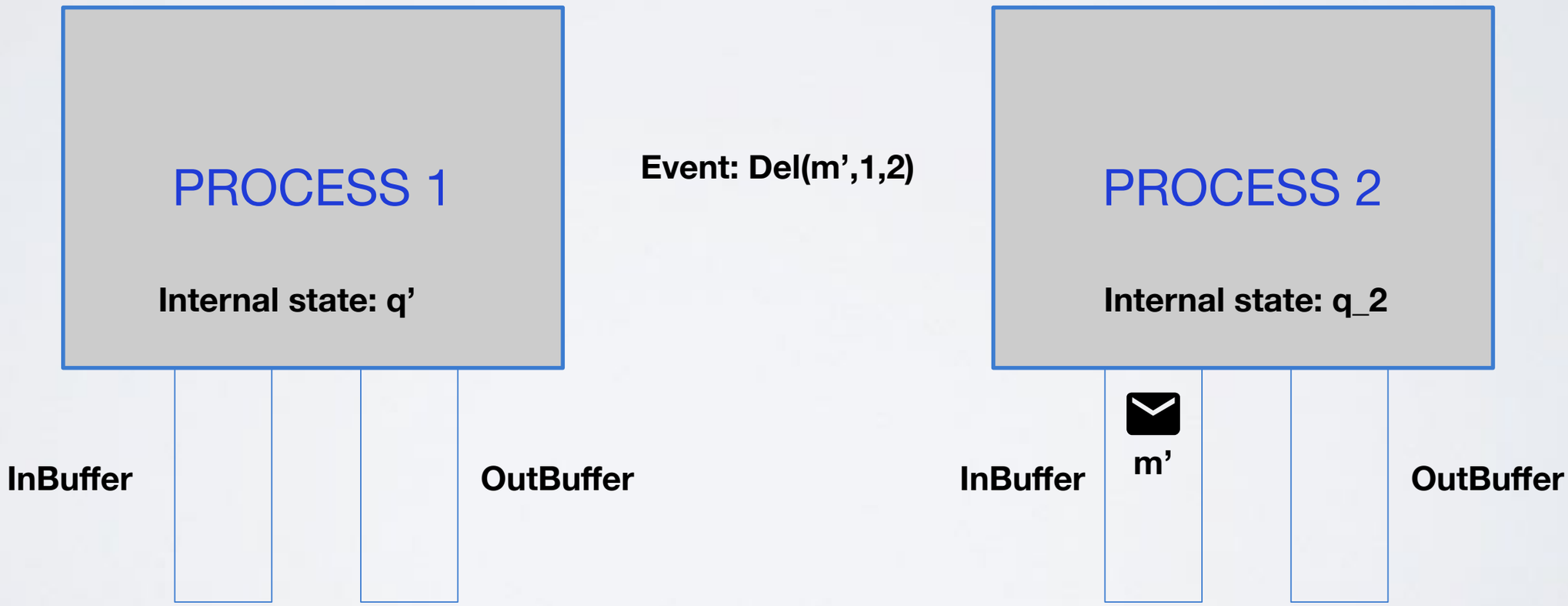
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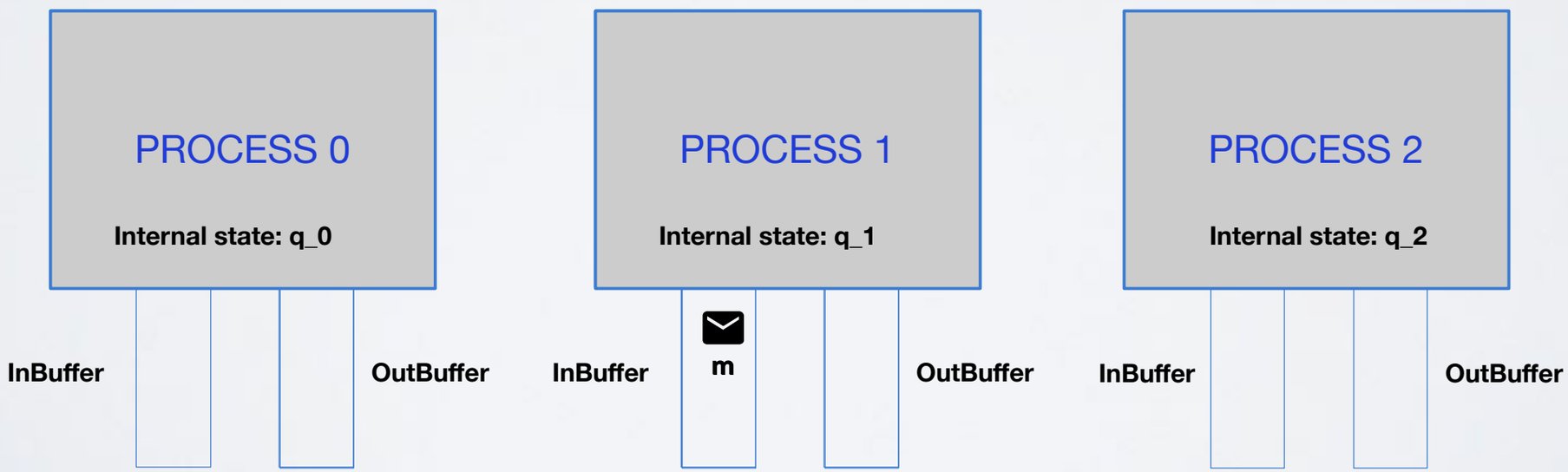


$$InBuf_2 = InBuf_2 \cup \{< 1, 2, m' >\}$$

EXECUTION

- A configuration C_t is a vector of n components. Component j indicates the state of process j (that is C_t[j] (q_i,InBuff_j,OutBuff_j)).

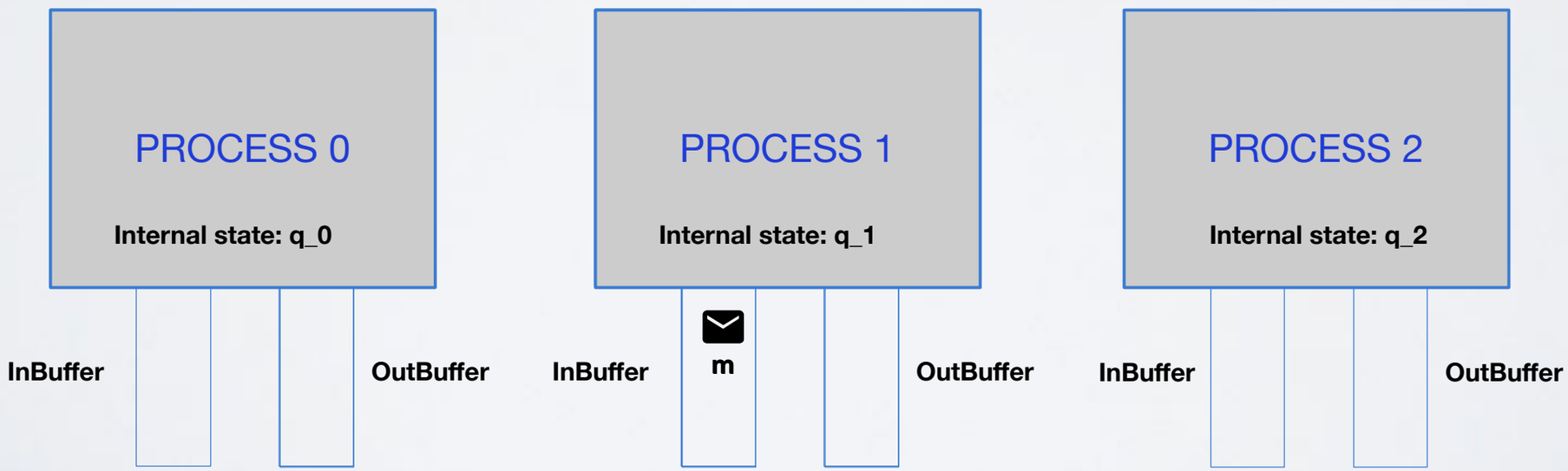
$$C_0 =< (q_0, \{\}, \{\}), (q_1, \{m\}, \{\}), (q_2, \{\}, \{\}) >$$



EXECUTION

- An event e is enabled in a configuration C if it can happen:
  - Del(m',0,2) is not enabled in C_0 because OutBuf_0 does not contain a message m'.
  - Exec(1) is enabled in C_0, as Exec(0) and Exec(2).

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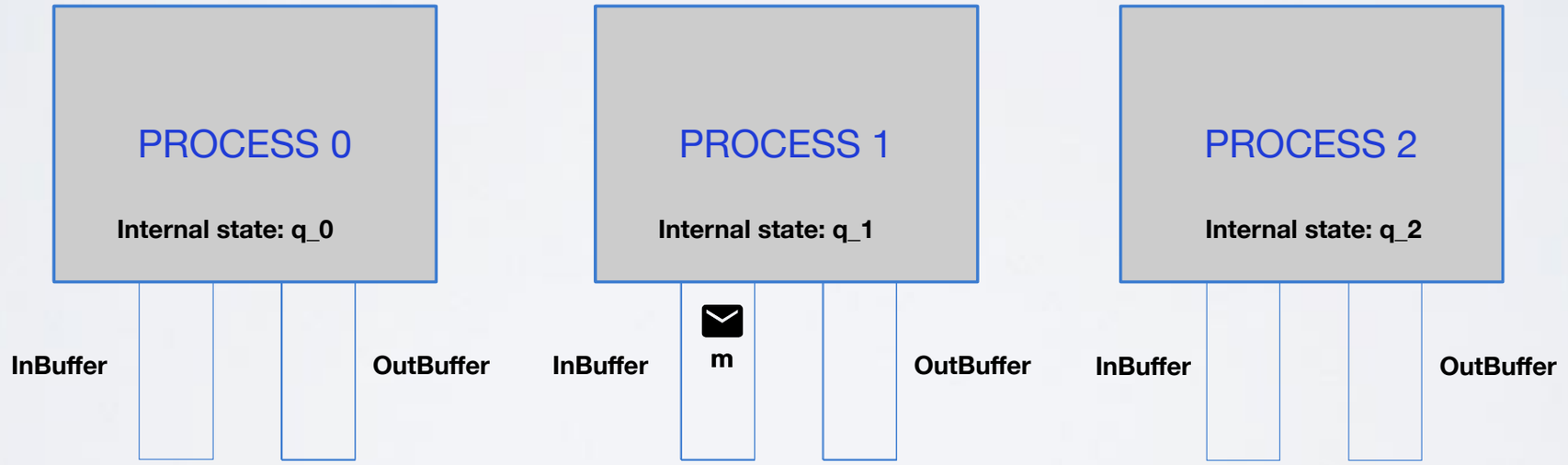


EXECUTION

- An execution is infinite sequence that alternates configurations and events: (C_0,e_0,C_1,e_1,C_2,e_2,….) such that each event e_{t} is enabled in configuration C_t and C_t is obtained by applying e_{t-1} to C_{t-1}.

$$\mathcal{E} : (C_0, e_0 = Exec(1), C_1, e_1 = Del(1, 2, m'), C_2, e_2 = Exec(2), C_3, e_3, C_4, \ldots)$$

C_0

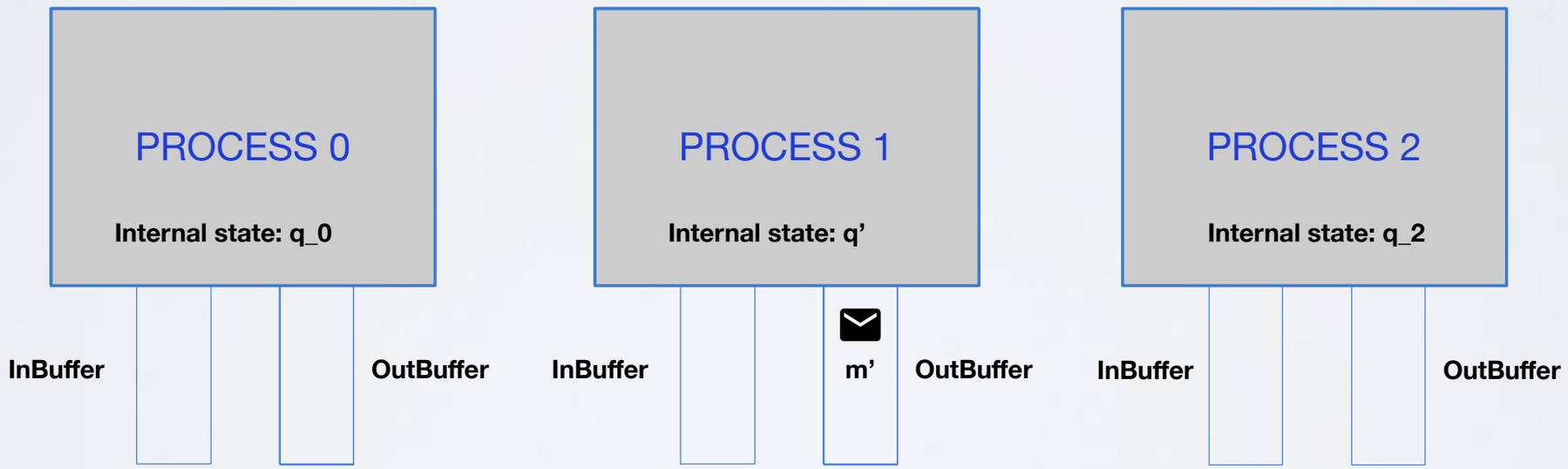


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C_1

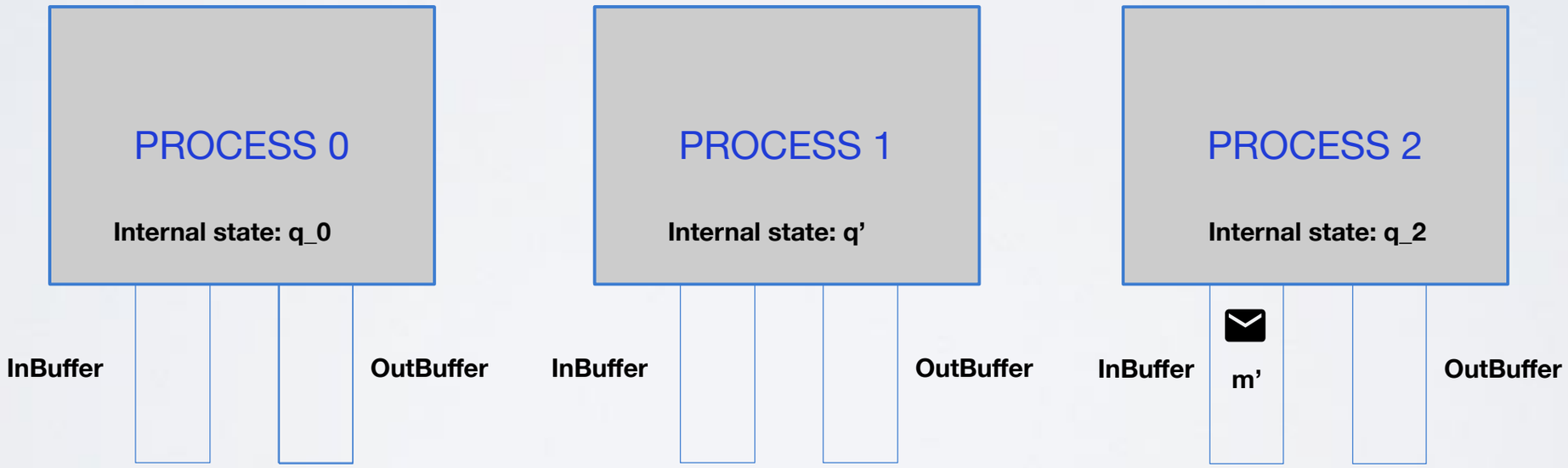


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C_2

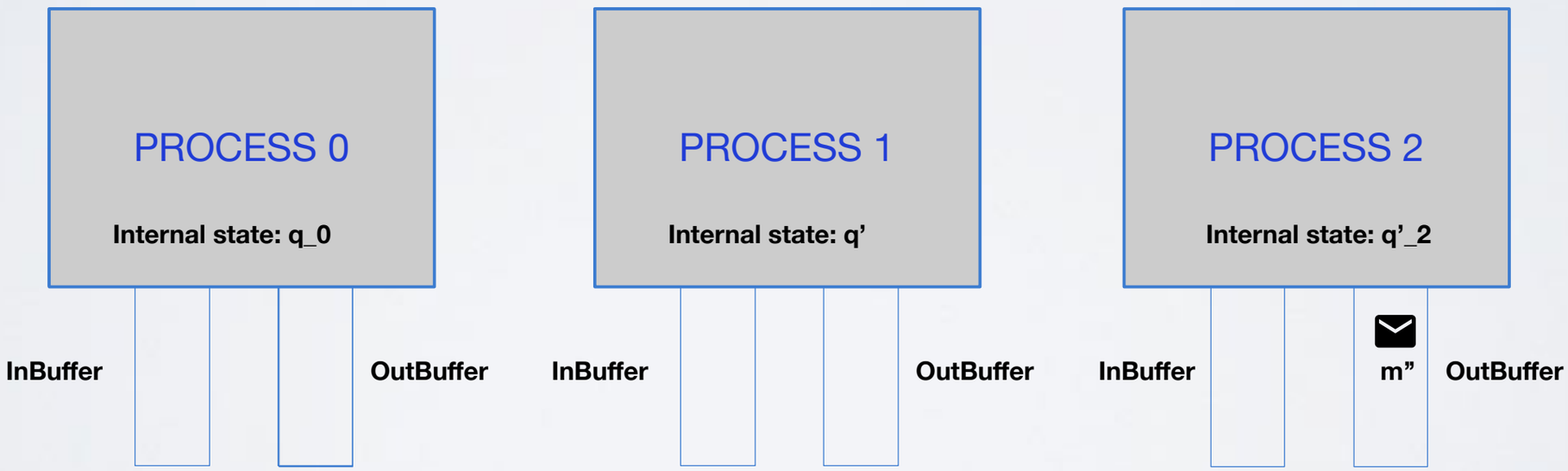


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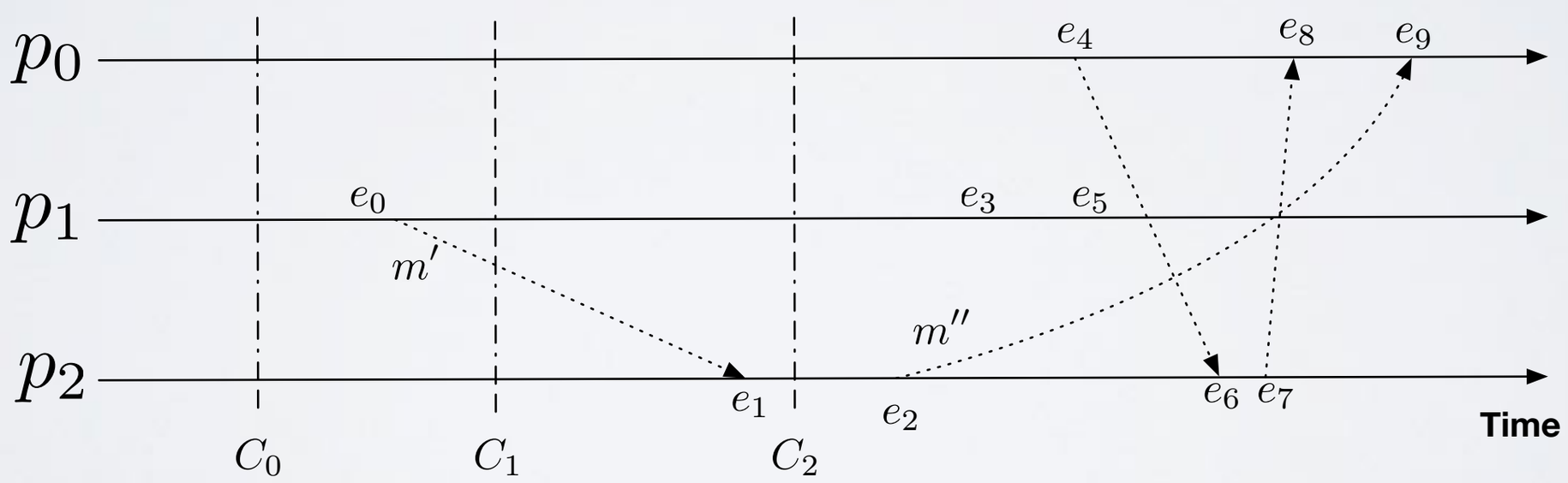
C_3



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FAIR EXECUTION

- An execution E is fair if each process p_i executes an infinite number of local computations (Exec(i) events are not finite) and each message m is eventually delivered (is not possible to stall forever a message: there must exists a Del(m,x,y)).

- We will always consider fair executions.

- Unfair executions brake any possible non-trivial algorithm (**why?**).

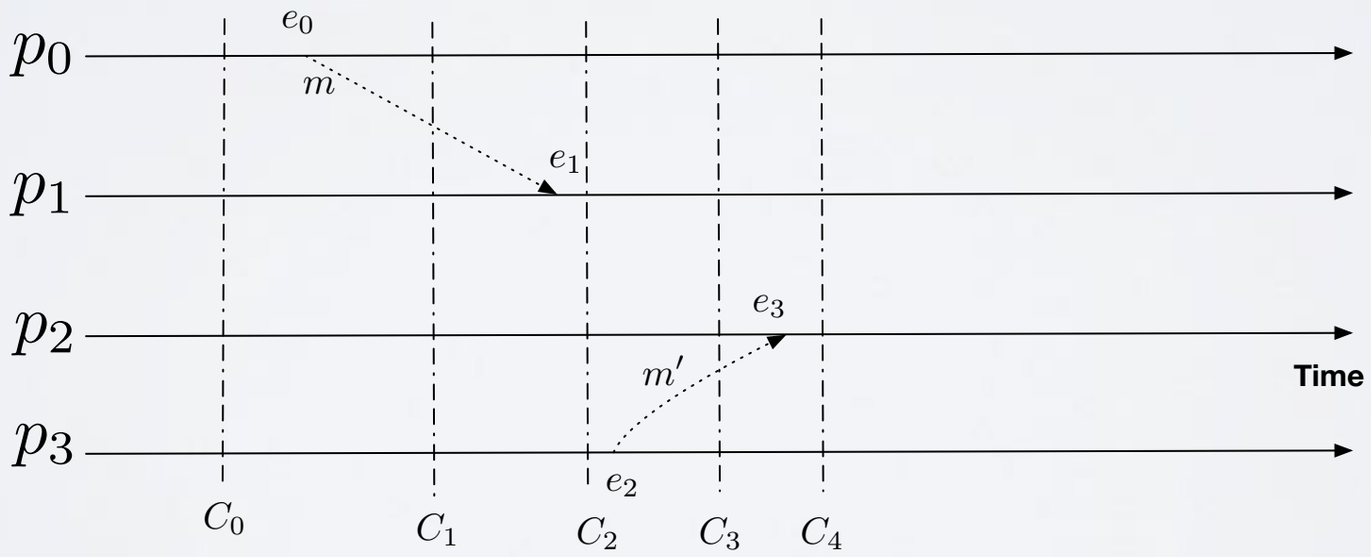
LOCAL EXECUTION (OR LOCAL VIEW)

- Given an execution E and a process p_j, we define has local execution (E|p_j) of p_j the subset of events in E that ``impact" p_j.

$$\mathcal{E} = (C_0, e_0 = Exec(0), C_1, e_1 = Del(0,1,m), C_2, e_2 = Exec(3), C_3, e_3 = Del(3,2,m'), ...)$$

$$\mathcal{E}|p_1 = (Del(0,1,m), ...)$$

$$\mathcal{E}|p_2 = (Del(3,2,m'), ...)$$

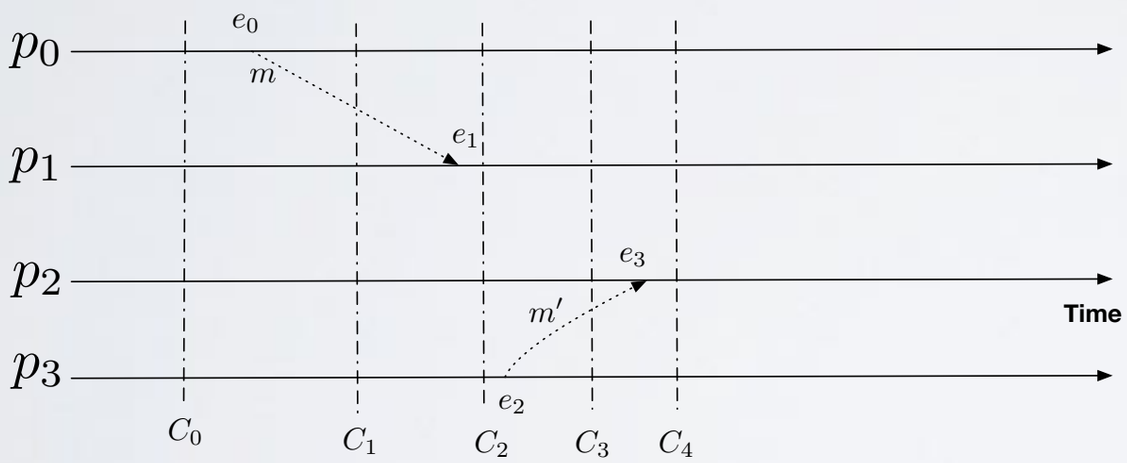


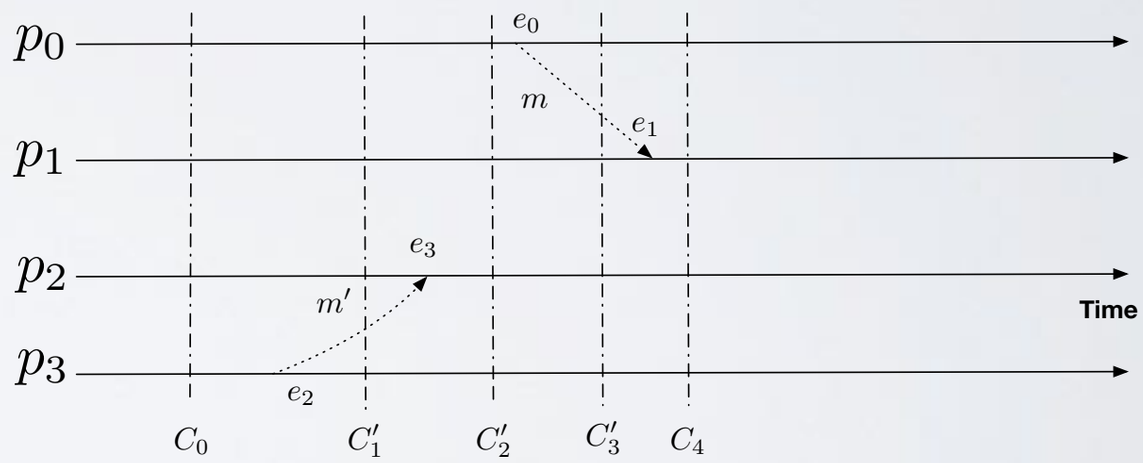
LOCAL EXECUTION

- Different executions could give the same local exec.:

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$$\mathcal{E}' = (C_0, e_2 = Exec(3), C_1', e_3 = Del(3,2,m'), C_2', e_0 = Exec(0), C_3', e_1 = Del(0,1,m), C_4, ...)$$

$$\mathcal{E}|p_1 = \mathcal{E}'|p_1 \text{ and } \mathcal{E}|p_2 = \mathcal{E}'|p_2$$



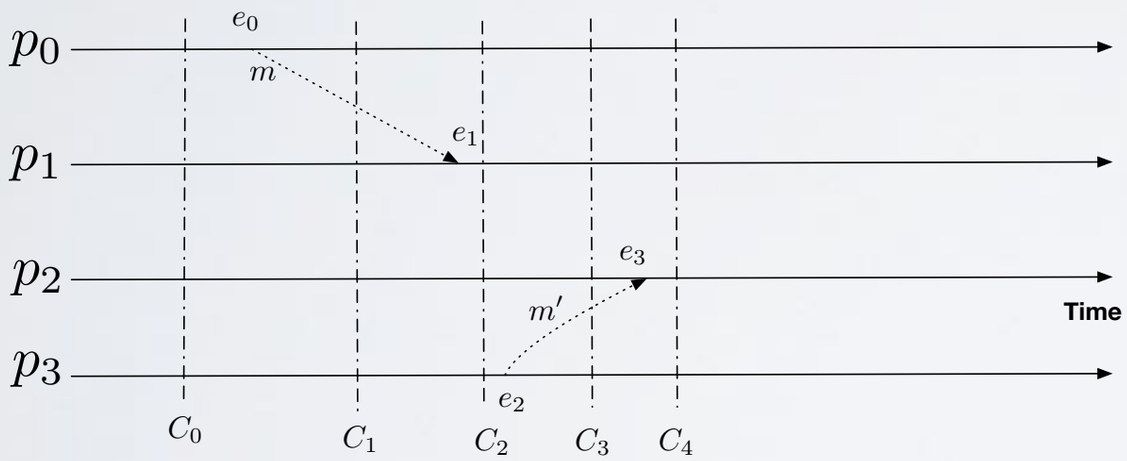


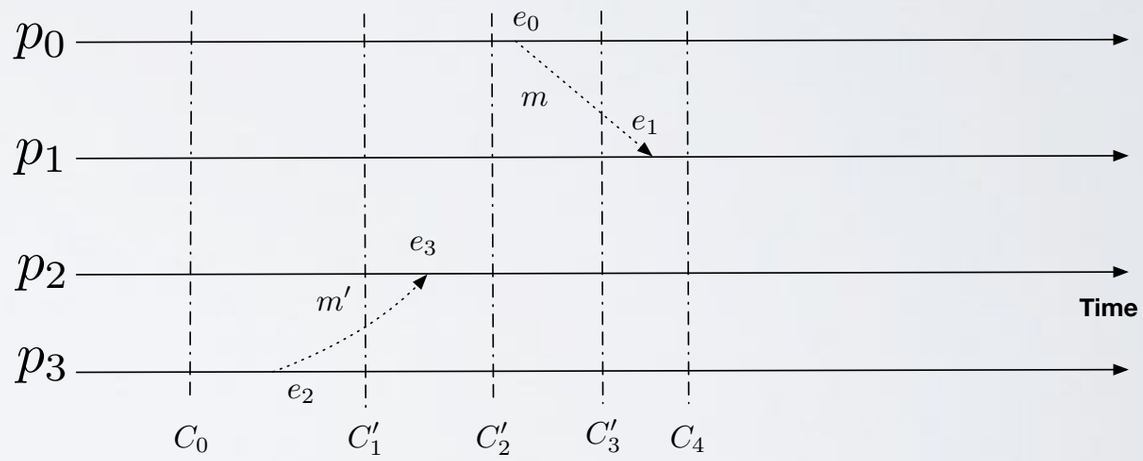
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$$\mathcal{E}|p_1 = \mathcal{E}'|p_1 \text{ and } \mathcal{E}|p_2 = \mathcal{E}'|p_2$$





- We say that p_1 (and p_2) cannot distinguish E from E'.

INDISTINGUISHABILITY

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- It is not hard to see that: $\forall p_j \in \Pi,\ \mathcal{E}|p_j = \mathcal{E}'|p_j$
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- **Theorem**: In the asynch. model there is no distributed algorithm capable of reconstructing the system execution.

SYNCHRONOUS VS ASYNCHRONOUS

SYNCHRONOUS

- In this course we will see three definitions of synchrony:

- Asynchronous system
- Eventually-synchronous (in module M2)
- Synchronous system

- A System is synchronous if there is fixed bound on the delay of messages, a fixed bound on the execution time of actions executed by processes, and a fixed bound between the execution of actions.

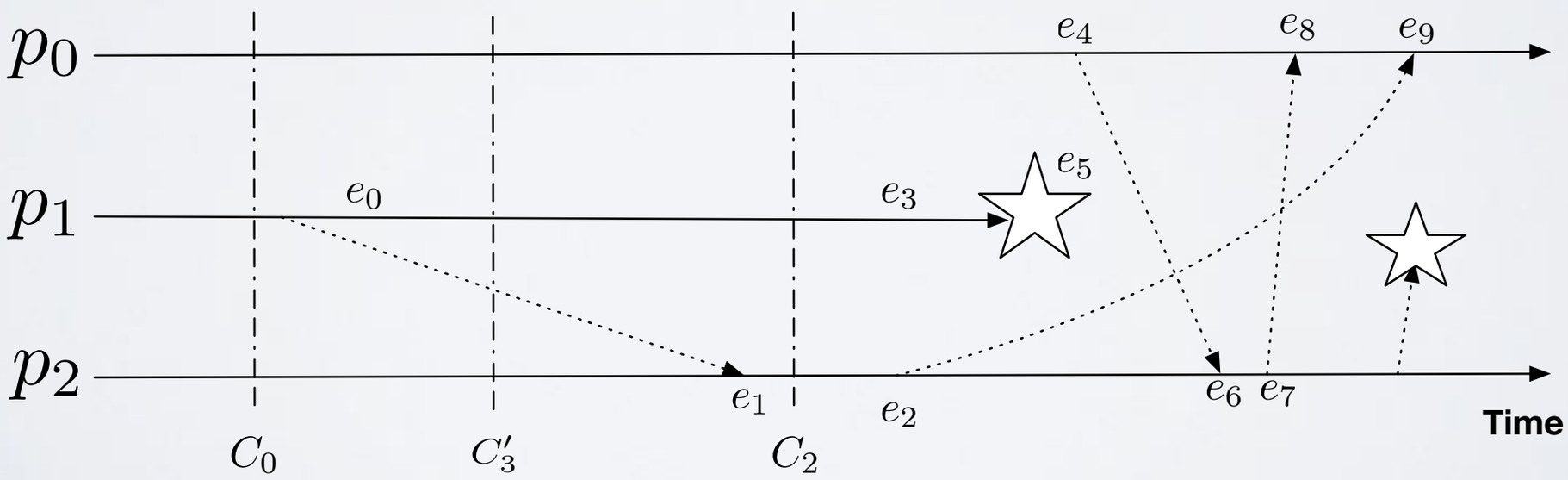
MODEL: FAILURES

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MODELS FOR PROCESS FAILURES

- Two main models:
  - Crash-Stop Failure
  - Byzantine Failure

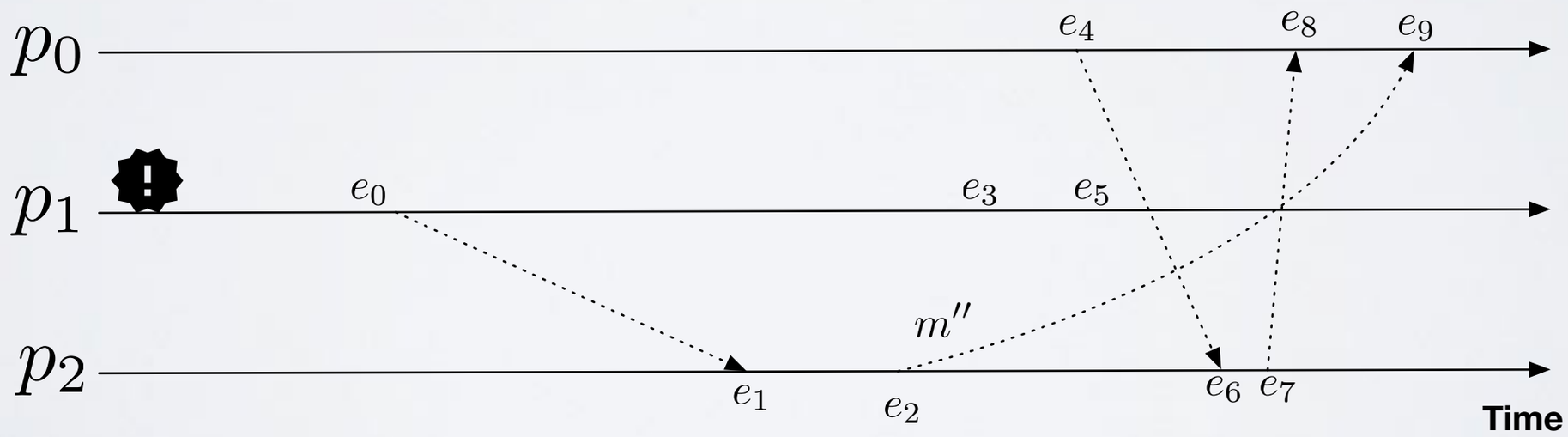
- Crash(p_j): after this event process p_j does not execute any local computation step (no Exec(j)).



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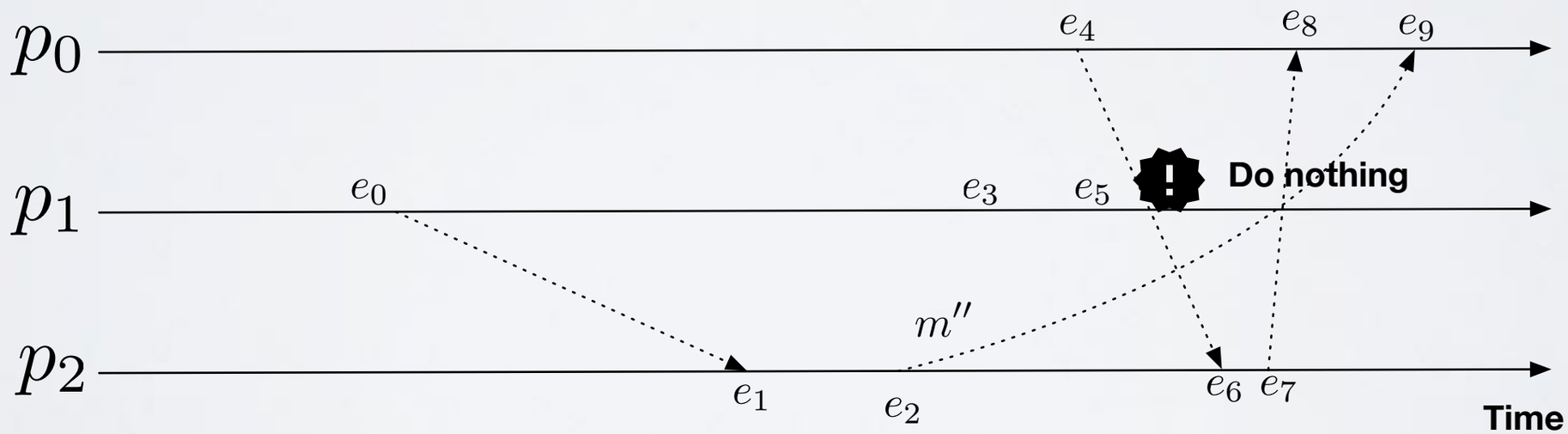
- Byz(p_j): after this event process p_j behaves in an arbitrary way (when we have Exec(j) we do not follow anymore the automaton of p_j).



MODELS FOR PROCESS FAILURES

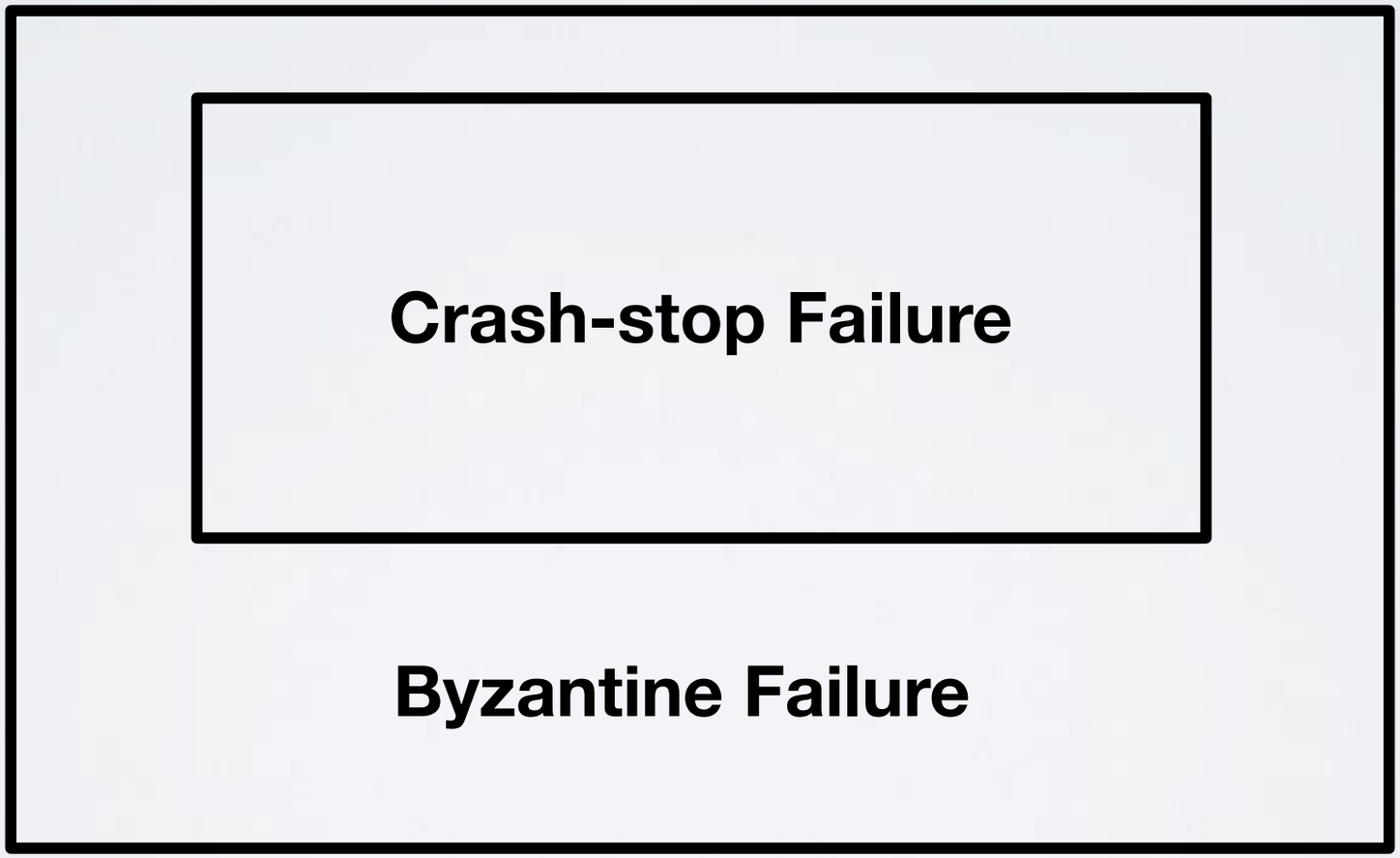
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MODELS FOR PROCESS FAILURES

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MODELS FOR PROCESS FAILURES

- Two main models:
  - Crash-Stop Failure
  - Byzantine Failure
- We say that a process is **correct** if it **does not experience a failure** (but we do not know who is correct).
- We indicate with **f** the **maximum number** of processes that can experience a failure event in an execution.
  - Usually we bound f. Example: f cannot be more than n-1.
  - We never do assumptions on when a failure event happens.

HOMEWORKS

- Model the loss of a message as an event.

ABSTRACTIONS, PSEUDO-CODE CONVENTIONS AND OUR FIRST ALGORITHM

ABSTRACTION AND PROPERTIES

THE ROAD TO BUILD A DISTRIBUTED ABSTRACTION

Step 1: define a system model

Step 2: formalize a problem/(object) with an abstraction

understand how to design an abstract formal object that captures the problem that you want to solve:

- Specify unambiguously the properties of your abstraction

Step 3: implements the abstraction with a distributed protocol

Step 4: prove that the protocol implements the abstraction

ABSTRACTING A REAL COMMUNICATION LINK

- Informal description: We have **a link** that loses messages with a certain probability pr, the channel can duplicate a message a finite number of times, and it does not create messages from thin air.

- A link exposes two events:
  - Requests: <Send | q,m> send a message m to process q.
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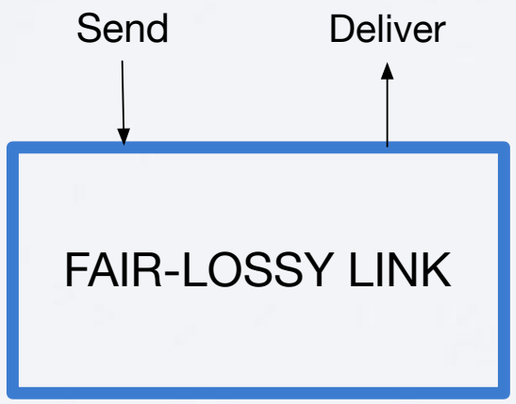
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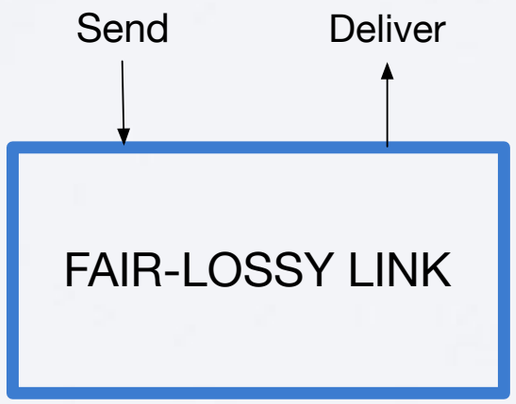
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LET US FOCUS ON PROPERTIES

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- All properties fall in two classes: ***Safety*** or ***Liveness***.
  - A **Safety property** is property that **if violated at a time t, it can never be satisfied after t.**
    - Formally, if a safety property is violated in execution E, there is a prefix E' of E such that any extension of E' also violates the property.

LET US FOCUS ON PROPERTIES

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
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- All properties fall in two classes: ***Safety*** or ***Liveness***.
  - A Safety property is property that if violated at a time t, it can never be satisfied after t.
    - Formally, if a safety property is violated in execution E, there is a prefix E' of E such that any extension of E' also violates the property.
    - Informally, **a safety property states that something bad cannot happen**.
    - Among FL1, FL2 and FL3 there is at least one safety property. Which one?

LET US FOCUS ON PROPERTIES

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.
  - **FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.**

- ***Safety:***
  - Once a process q delivers a message m that was not sent by anyone FL3 is violated forever.
    - E=(e_1,e_2,e_3,e_4,e_5,……)
    - E'=(e_1,e_2,e_3,e_4)

LET US FOCUS ON PROPERTIES

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- All properties fall in two classes: ***Safety*** or ***Liveness***.
  - A Liveness property is property that cannot be violated in finite executions:
    - Formally, given any finite execution E that does not satisfy a liveness property there is an extension of E that satisfy it.
    - Informally, a liveness property specify that something good will happen.
    - Among FL1, FL2 and FL3 there is at least one liveness property. Which one?

LET US FOCUS ON PROPERTIES

- Properties:
  - **FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.**
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- ***Liveness***.
  - If p sent a message m, and q has not delivered it at time t, there is always hope that it can deliver it at time t+\Delta.
    - E=(e_1,e_2,e_3,e_4,e_5)
    - E'=(e_1,e_2,e_3,e_4,e_5, del(p,q,m))

LET US FOCUS ON PROPERTIES

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
  - **FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.**
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.
- ***?***

LET US FOCUS ON PROPERTIES

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
  - **FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.**
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- ***Liveness.***
  - It cannot be violated in finite execution.
    - E=(del(p,q,m),del(p,q,m),del(p,q,m),…,del(p,q,m))
    - E'=(del(p,q,m),del(p,q,m),del(p,q,m),…,del(p,q,m),e_1,e_2,e_3,….)
      - Such that no e_j is del(p,q,m).

LET US FOCUS ON PROPERTIES

- Property:
  - **FL2: (At-most7 duplication)** If a correct process p sends m a finite number of times to q, then q cannot deliver m **more than 7 times**.
- ***Liveness?***

LET US FOCUS ON PROPERTIES

- Property:
  - **FL2: (At-most7 duplication)** If a correct process p sends m a finite number of times to q, then q cannot deliver m **more than 7 times**.
- ***Liveness?***
  - E=(del(p,q,m),del(p,q,m),del(p,q,m),del(p,q,m),del(p,q,m),del(p,q,m),del(p,q,m),del(p,q,m))
    - E cannot be extended, it always violates At-most7 duplication.
  - At-most7 duplication is a **safety** property.

OTHER PROPERTIES

- Mutual Exclusion: if a process p is granted a resource r at time t, then no other process q is granted r at t.
- No-deadlock: if r is not already granted, eventually someone get a grant on resource r.
- No-Starvation: if a process p request a grant on resource r, it will eventually get it.

BADLY WRITTEN PROPERTIES

- If process p sends a message m to q, then q will eventually deliver it and this deliver is unique.
  - It is mixing two aspects:
    - A liveness: q will eventually deliver it
    - A safety: the deliver is unique

BADLY WRITTEN PROPERTIES

- If process p sends a message m to q, then q will eventually deliver it and this deliver is unique.
  - It is mixing two aspects:
    - A liveness: q will eventually deliver it
    - A safety: the deliver is unique
  - It should be decomposed in two properties:
    - If p sends a message m to q, then q eventually delivers it.
    - If p sends a message m to q, then m is delivered at most once.

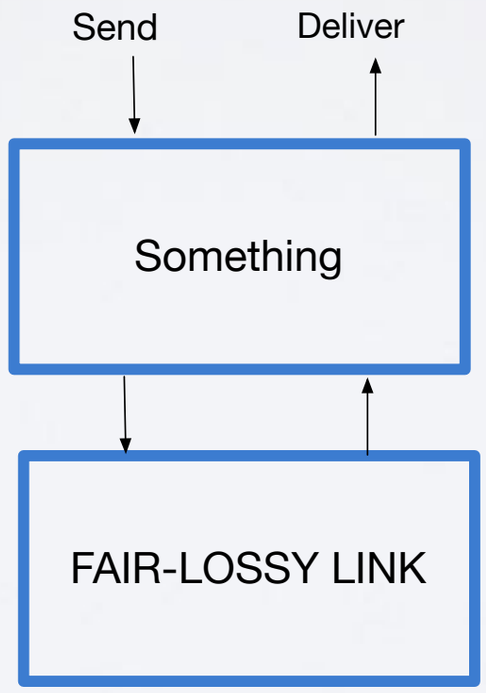
HOMEWORKS

- Safety or liveness?
  - Fairness of an execution.
  - (At most-f failures) An execution E can contain at most f failures.
  - (Eventual recovery) If a process p experience a crash(p), then it eventually experiences also a recovery(p) (this event starts again process p).
  - (Time-bounded recovery) If a process p experiences a recovery(p), then this recovery happens at most k events after its failure.
- Write three new safety properties and three new liveness properties.
- Does Time-bounded recovery implies Eventual recovery?
- How would you improve the writing of this property:
  - If a process p outputs a set of integer O_p, then any other process q outputs also outputs a set O_q. For any two processes z and t we have either O_z is a subset of O_t or that O_t is a subset of O_z.

ALGORITHMS: COMMUNICATION LINKS

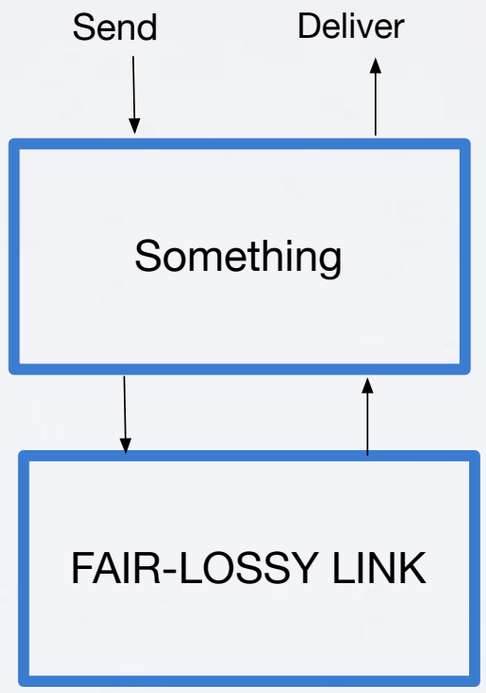
LET US GO BACK ON OUR FL-LINK

- Properties:
  - FL1: (Fair-loss) If a correct process p sends infinitely often m to a process q, then q delivers m an infinite number of times.
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, then q cannot deliver m an infinite number of times.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.



LET US GO BACK ON OUR FL-LINK

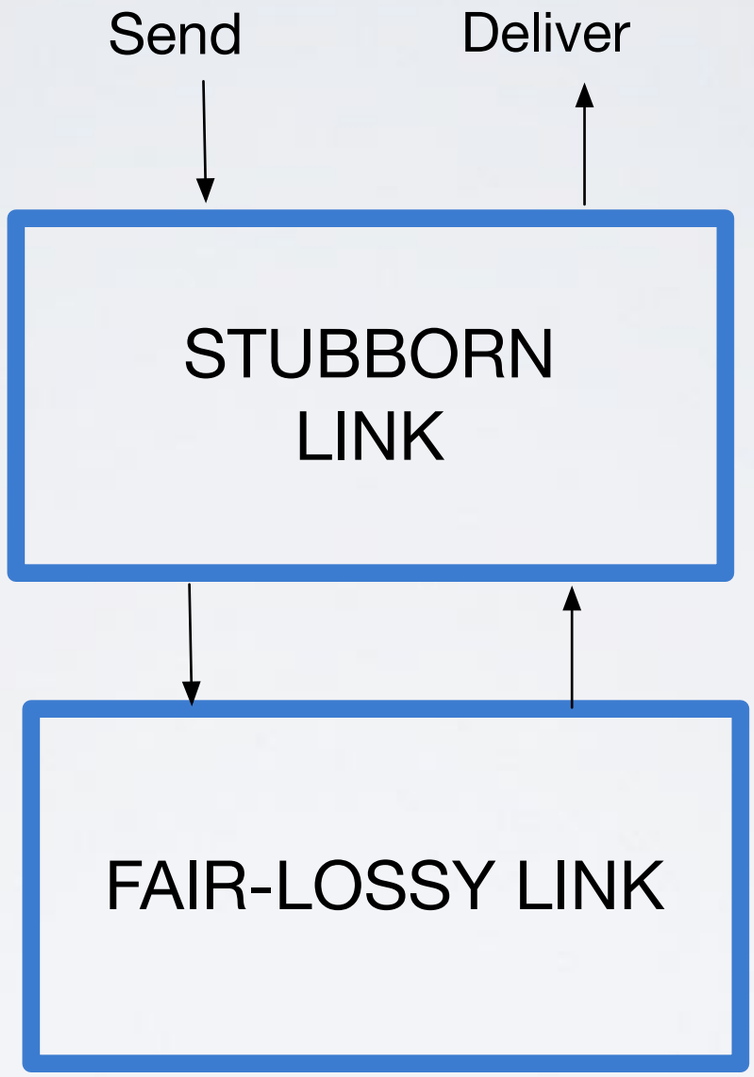
- Properties:
  - FL1: (Fair-loss) If a correct process **p sends infinitely often m** to a process q, then q delivers m an infinite number of times.
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, **then q cannot deliver m an infinite number of times**.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.



LET US GO BACK ON OUR FL-LINK

- Let us fix:
  - FL1: (Fair-loss) If a correct process **p sends infinitely often m** to a process q, then q delivers m an infinite number of times.
  - FL2: (Finite duplication) If a correct process p sends m a finite number of times to q, **then q cannot deliver m an infinite number of times**.
- In:
  - SL1: (Stubborn-delivery) If a correct process p sends m to q, then q delivers m an infinite number of times.
- While we keep:
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

OUR FIRST ALGORITHM



COMPOSITION MODEL

- The protocols we will consider in this course are presented in pseudo-code
- The pseudo code reflects a reactive computing model where
  - components of the same process communicate by exchanging events
  - the algorithm is described as a set of event handlers
  - handlers react to incoming events and possibly trigger new events.
  - Handlers are atomic. They can only be interrupted in case of a crash.

STUBBORN

- Events:
  - Requests: <Send | q,m> send a message m to process q.
  - Indication: <Deliver, p,m> delivers message m from process p.
- Properties:
  - SL1: (Stubborn-delivery) If a correct process p sends m to q, then q delivers m an infinite number of times.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

STUBBORN

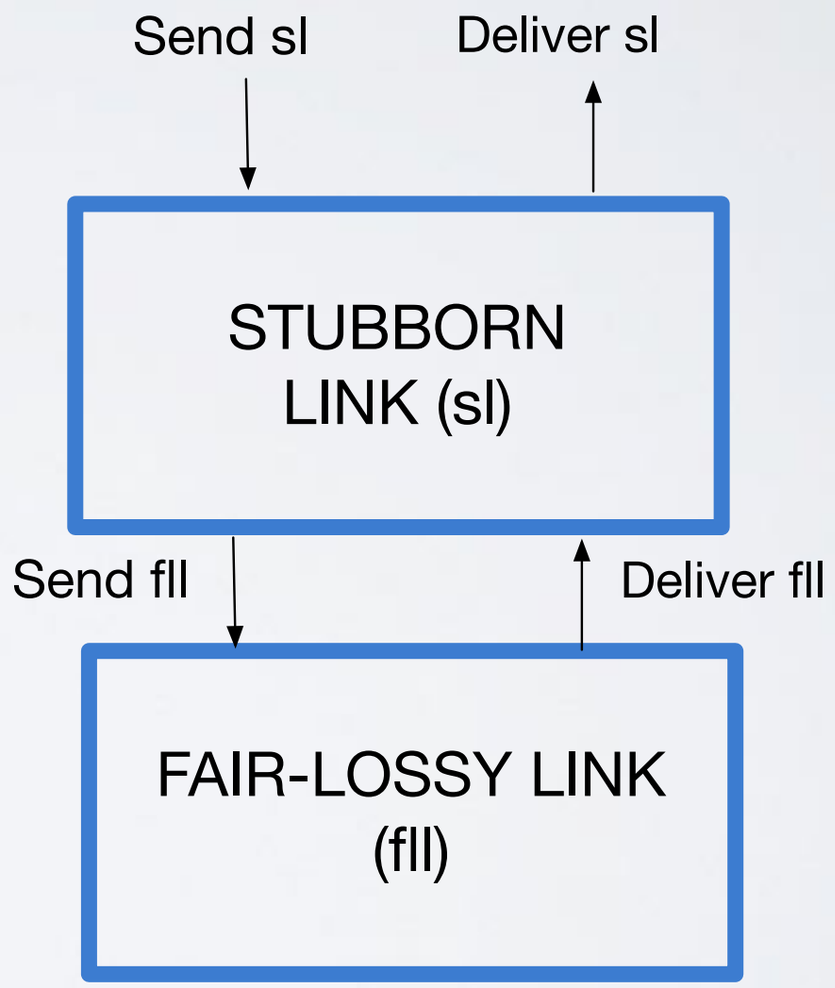
- Properties:
  - SL1: (Stubborn-delivery) If a correct process p sends m to q, then q delivers m an infinite number of times.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

upon event $\langle\ sl, Init\ \rangle$ **do**
    $sent := \emptyset$;
    $starttimer(\Delta)$;

upon event $\langle\ Timeout\ \rangle$ **do**
    forall $(q, m) \in sent$ **do**
        trigger $\langle\ fll, Send \mid q, m\ \rangle$;
    $starttimer(\Delta)$;

upon event $\langle\ sl, Send \mid q, m\ \rangle$ **do**
    trigger $\langle\ fll, Send \mid q, m\ \rangle$;
    $sent := sent \cup \{(q, m)\}$;

upon event $\langle\ fll, Deliver \mid p, m\ \rangle$ **do**
    trigger $\langle\ sl, Deliver \mid p, m\ \rangle$;



STUBBORN FORMAL PROOF

- FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- Proof. By contradiction.
- Suppose process q executing our algorithm receives message m that was not sent by p.
- Fact 1: If q delivers a message, then it delivers here:

upon event $\langle$ *fll*, *Deliver* | *p*, *m* $\rangle$ **do**
trigger $\langle$ *sl*, *Deliver* | *p*, *m* $\rangle$;

STUBBORN FORMAL PROOF

- FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- Proof. By contradiction.
- Suppose process q executing our algorithm receives message m that was not sent by p.
- Fact 1: If q delivers a message, then it delivers here:

upon event ⟨ *fll*, *Deliver* | *p*, *m* ⟩ **do**
trigger ⟨ *sl*, *Deliver* | *p*, *m* ⟩;

- Fact 1 implies that fll, is delivering a message that was not sent by p. This implies that fll is not fair-lossy. This contradicts our hypothesis: fll is fair-lossy.

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STUBBORN FORMAL PROOF

- SL1: (Stubborn-delivery) If a correct process p sends m to q, then q delivers m an infinite number of times.
- Proof. By contradiction. Suppose q delivers m a finite number of times.
- Fact 1: p sends m on fll an infinite number of times:

```
upon event ⟨ Timeout ⟩ do
    forall (q, m) ∈ sent do
        trigger ⟨ fll, Send | q, m ⟩;
    starttimer(Δ);

upon event ⟨ sl, Send | q, m ⟩ do
    trigger ⟨ fll, Send | q, m ⟩;
    sent := sent ∪ {(q, m)};
```

- Fact 2: if q ``stubborn delivers" m a finite number of times, then fll delivered m a finite number of times.

```
upon event ⟨ fll, Deliver | p, m ⟩ do
    trigger ⟨ sl, Deliver | p, m ⟩;
```

IMPROVING STUBBORN

- Properties:
  - SL1: (Stubborn-delivery) If a correct process p sends m to q, then q **delivers m an infinite number of times**.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.
- In:
  - PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.
  - PL2: (No duplication) A message is delivered at most once.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

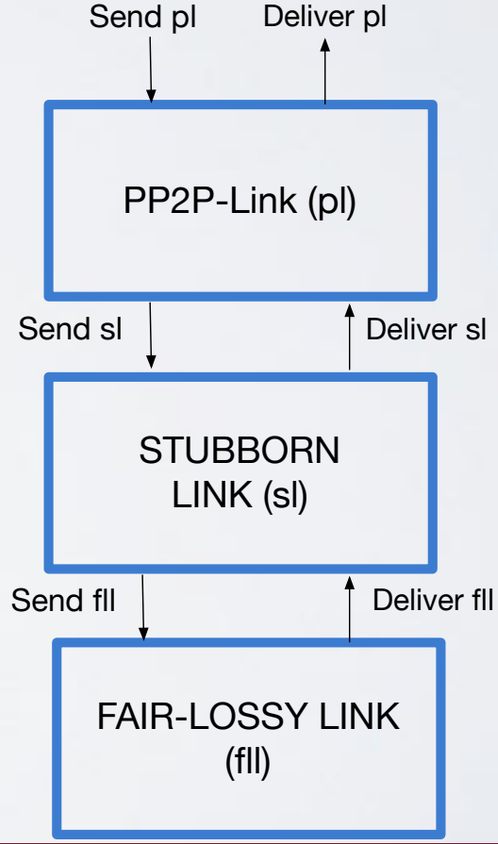
PERFECT P2P LINK

- Properties:
  - PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.
  - PL2: (No duplication) A message is delivered at most once.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

upon event $\langle\ pl, Init\ \rangle$ **do**
    $delivered := \emptyset;$

upon event $\langle\ pl, Send \mid q, m\ \rangle$ **do**
    trigger $\langle\ sl, Send \mid q, m\ \rangle;$

upon event $\langle\ sl, Deliver \mid p, m\ \rangle$ **do**
    if $m \notin delivered$ **then**
        $delivered := delivered \cup \{m\};$
        trigger $\langle\ pl, Deliver \mid p, m\ \rangle;$



PERFECT P2P LINK - FORMAL PROOF

- FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.

- Proof. By contradiction.
- Suppose process q executing our algorithm receives message m that was not sent by p.
- Fact 1: If q delivers a message, then it delivers here:

upon event $\langle\ sl, Deliver \mid p, m\ \rangle$ **do**
  if $m \notin delivered$ **then**
    $delivered := delivered \cup \{m\}$;
    trigger $\langle\ pl, Deliver \mid p, m\ \rangle$;

- This implies that sl delivered a message that was not created. Violates the hypothesis that sl is a stubborn.

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PERFECT P2P LINK - FORMAL PROOF

- PL2: (No duplication) A message is delivered at most once
- Fact 1: The pp2p-delivery of a message is ``guarded'' by an if m \in delivered:

upon event $\langle$ *sl*, *Deliver* | *p*, *m* $\rangle$ **do**
  if $m \notin$ *delivered* **then**
    delivered := *delivered* $\cup \{m\}$;
    trigger $\langle$ *pl*, *Deliver* | *p*, *m* $\rangle$;

- Suppose m is delivered twice, at time t and t' (with t< t').
- At time t the delivery handler is executed. Since the handler is atomic we have that delivered:=delivered \cup {m} is executed before t' Therefore, at t' m is in delivered, this contradict the fact that trigger <pl, deliver | P, m> is executed at (or after) time t'.

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PERFECT P2P LINK - FORMAL PROOF

- PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.

- Suppose, p sends m and q does not deliver it. There could be two reasons for q to not deliver m:
- Reason 1: m is in delivered when the delivery handler is executed:

upon event $\langle\ sl, Deliver \mid p, m\ \rangle$ **do**
  if $m \notin delivered$ **then**
    $delivered := delivered \cup \{m\}$;
    trigger $\langle\ pl, Deliver \mid p, m\ \rangle$;

- If m is in delivered then, q eventually will execute **trigger** <pl, Deliver| p, m>. This contradicts the fact that q does not deliver m.

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PERFECT P2P LINK - FORMAL PROOF

- PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.

- Suppose, p sends m and q does not deliver it. There could be two reasons for q to not deliver m:
- Reason 2: the delivery handler is never triggered with <p,m>

upon event $\langle\ sl, Deliver \mid p, m\ \rangle$ **do**
  if $m \notin delivered$ **then**
    $delivered := delivered \cup \{m\}$;
    trigger $\langle\ pl, Deliver \mid p, m\ \rangle$;

- This means that sl is not stubborn. Violating our hypothesis.

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HOMEWORKS 1

- Show that our stubborn algorithm does not work if we change the first property in:
  - SL1: (Stubborn-delivery) If a process p sends m to q, then q delivers m an infinite number of times. {Pay attention something is missing}.
- Show that there is no algorithm that satisfies the property above using a fair-lossy link.
- *Suppose you can use only a fixed amount k of memory (see it as having the possibility to store at most k different messages), could you implement a Perfect p2p link on a fair-lossy?

HOMEWORKS 1

- Show that our stubborn algorithm does not work if we change the first property in:
  - SL1: (Stubborn-delivery) If a process p sends m to q, then q delivers m an infinite number of times. {Pay attention something is missing}.
  - The property does not specify the correctness of p. This means that even if p crashes, then q has to deliver m (at least once).
  - If p crashes, then it cannot send an infinite number of messages on fll.
  - If p sends a finite number of messages on fll, then it could be that no message is delivered to q.

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HOMEWORKS 1

- Show that our stubborn algorithm does not work if we change the first property in:
  - SL1: (Stubborn-delivery) If a process p sends m to q, then q delivers m an infinite number of times. {Pay attention something is missing}.
- Show that there is no algorithm that satisfies the property above using a fair-lossy link.
- The same argument shows that no algorithm could satisfy the property above on a fair-lossy link.

REASONING TIME…

- Recall the following property of a perfect p2p link:
  - PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.

- Assume a system with crash failure, is there something wrong?

REASONING TIME…

- PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.

- Suppose, p sends m and q does not deliver it. There could be two reasons for q to not deliver m:
- Reason 1: m is in delivered when the delivery handler is executed:

upon event $\langle\ sl, Deliver \mid p, m\ \rangle$ **do**
  if $m \notin delivered$ **then**
    $delivered := delivered \cup \{m\}$;
    trigger $\langle\ pl, Deliver \mid p, m\ \rangle$;

- If m is in delivered then, q eventually will execute **trigger** <pl, Deliver| p, m>. This contradicts the fact that q does not deliver m.

REASONING TIME…

- Recall the following property of a perfect p2p link:
  - PL1: (Reliable delivery) If a correct process p sends m **to q and q is correct**, then q eventually delivers m.

HOMEWORKS 2

- One of your friends, Caio, is really excited about its new discovery. Caio claims to have implemented a stubborn over a fair-lossy without the need of any internal timer or infinite loop. It shows you this algorithm:

upon event INIT
  $set = \emptyset$

upon event SL, SEND($< m, q >$)
  $set = set \cup \{< m, q >\}$
  for all $< x, y > \in set$ **do**
    invoke fl Send($< x, y >$)

upon event FL, DELIVER($< m, q >$)
  invoke sl Deliver($< m, q >$)

What is wrong with his algorithm? In which executions does it work? Could you find an argument that convinces Caio on the impossibility of implementing a stubborn (and thus a perfect link) over a fair-lossy without using a timer or an infinite loop?

(note that an infinite loop is actually a timer)

HOMEWORKS 2

What is wrong with his algorithm? In which executions does it work? Could you find an argument that convinces Caio on the impossibility of implementing a stubborn (and thus a perfect link) over a fair-lossy without using a timer or an infinite loop?

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    invoke fl Send($< x, y >$)

upon event FL, DELIVER($< m, q >$)
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HOMEWORKS 2

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(note that an infinite loop is actually a timer)

upon event INIT
  $set = \emptyset$

upon event SL, SEND($< m, q >$)
  $set = set \cup \{< m, q >\}$
  for all $< x, y > \in set$ **do**
    invoke fl Send($< x, y >$)

upon event FL, DELIVER($< m, q >$)
  invoke sl Deliver($< m, q >$)

The algorithm does not implement PL1: (Reliable delivery). Suppose, p stubborn sends message m, and then it stops sending messages. Then the fair-lossy send is triggered only once. This implies that q could not deliver m.

HOMEWORKS 2

What is wrong with his algorithm? **In which executions does it work?** Could you find an argument that convinces Caio on the impossibility of implementing a stubborn (and thus a perfect link) over a fair-lossy without using a timer or an infinite loop?

(note that an infinite loop is actually a timer)

upon event INIT
  $set = \emptyset$

upon event SL, SEND($< m, q >$)
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HOMEWORKS 2

What is wrong with his algorithm? **In which executions does it work?** Could you find an argument that convinces Caio on the impossibility of implementing a stubborn (and thus a perfect link) over a fair-lossy without using a timer or an infinite loop?

(note that an infinite loop is actually a timer)

upon event INIT
  $set = \emptyset$

upon event SL, SEND($< m, q >$)
  $set = set \cup \{< m, q >\}$
  for all $< x, y > \in set$ **do**
    invoke fl Send($< x, y >$)

upon event FL, DELIVER($< m, q >$)
  invoke sl Deliver($< m, q >$)

The algorithm works only if all messages in set are fll sent an infinite number of times. Therefore, it only works in executions in which p stubborn sends an infinite number of messages.

HOMEWORKS 2

What is wrong with his algorithm? In which executions does it work? **Could you find an argument** that convinces Caio **on the impossibility of implementing a stubborn** (and thus a perfect link) **over a fair-lossy without using a timer or an infinite loop?**

(note that an infinite loop is actually a timer)

HOMEWORKS 2

What is wrong with his algorithm? In which executions does it work? **Could you find an argument** that convinces Caio **on the impossibility of implementing a stubborn** (and thus a perfect link) **over a fair-lossy without using a timer or an infinite loop?**

(note that an infinite loop is actually a timer)

Proof: Suppose you can, then in your algorithm you must have an infinite number of internal events in which you send message m over the fair-lossy link.

If you do not have an infinite loop (or a timer) and p does not receive an infinite number of input events, then it means that after a certain number of local execution steps, p will never change its internal state neither execute any actions.

Therefore, there exists at least an execution in which p executes a finite number of actions, in particular p sends m on the fair-lossy a finite number of times. This implies that is impossible for the algorithm to ensure the delivery of m by q

HOMEWORKS BIS

From the question of one of you colleagues:

- We have seen that the fairness property requires a process to be scheduled forever, so if a process crashes the execution is not fair. The reason is that a failed process will never be scheduled again for execution.

How this is **consistent** with the definition of fairness?

- The reason is that the definition of Fairness has been given for a model without failures! In case of processes failures, we have to modify the definition of fairness. That becomes:

HOMEWORKS BIS

From the question of one of you colleagues:

- We have seen that the fairness property requires a process to be scheduled forever, so if a process crashes the execution is not fair. The reason is that a failed process will never be scheduled again for execution.

How this is **consistent** with the definition of fairness?

- An execution E is fair if each **correct** process p_i executes an infinite number of local computations (Exec(i) events are not finite) and each message m sent **by a correct process or to a correct process** is eventually delivered.

LINKS

- You built a COUNTING SERVER that communicates with you using our Perfect-p2p algorithm over a stubborn link. The server stores a variable v (initially equal to 0) and it exposes three operations:
  - INC: that increases of 1 the value of v.
  - DEC: that decreases of 1 the value of v.
  - GET: that invokes <pl, Send | v, client> (client is your name as a process).

The server executes an operation when it delivers the corresponding message. As example, the increase happens when the server sees the event <pl, Deliver | INC,client>.

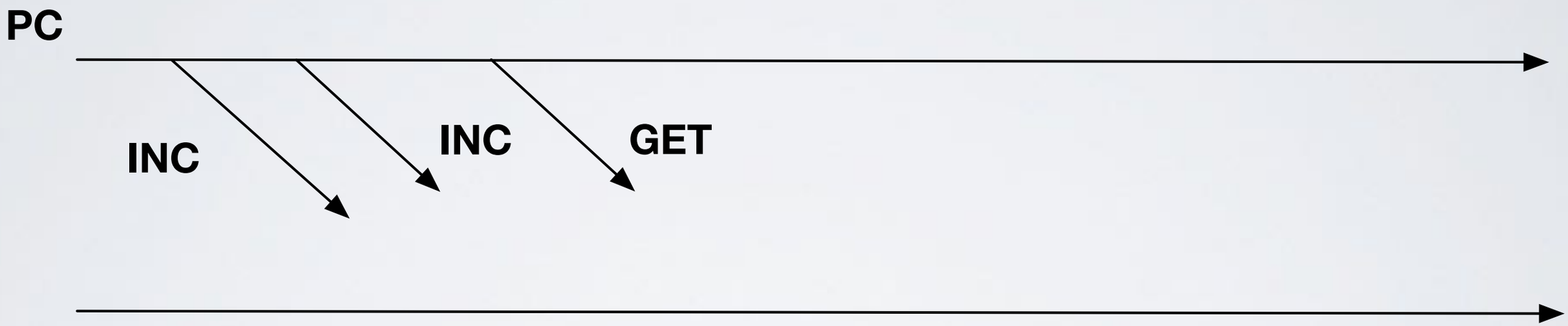
You act as a client and you invoke exactly three operations in the following order: <pl, Send| INC, server>,<pl, Send| INC, server>,<pl, Send| GET, server>.

You are really puzzled when, after a while, on your client you see <pl, Deliver| 1, server>. Startled, you asked again <pl, Send| GET, server>. Unfortunately, you are still waiting for an answer.

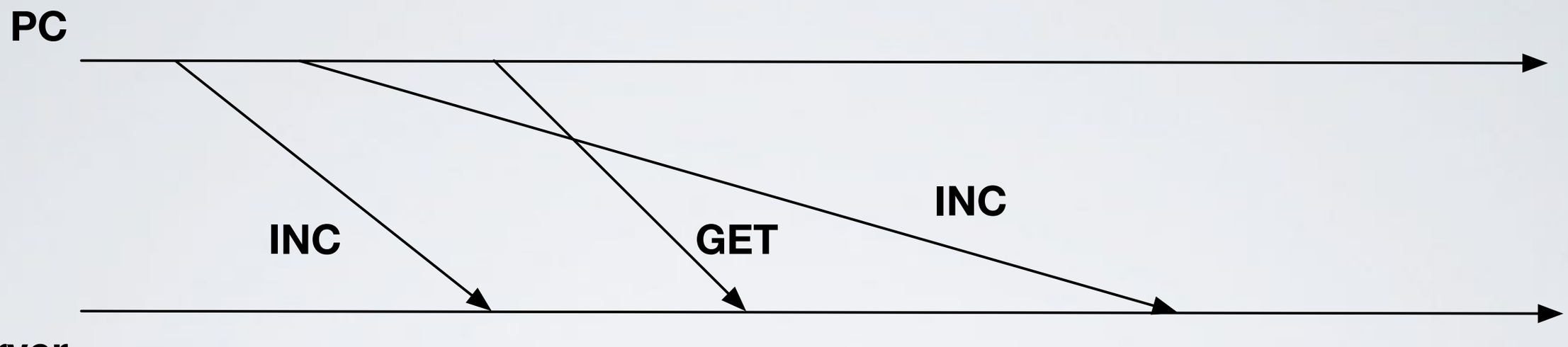
Questions: What is happening? Is there any way to fix it?

Note that, there are two possible explanations, try to find both. (Hint: you can see one of the explanation by answering to the question: How much will you wait for an answer?)

LINKS



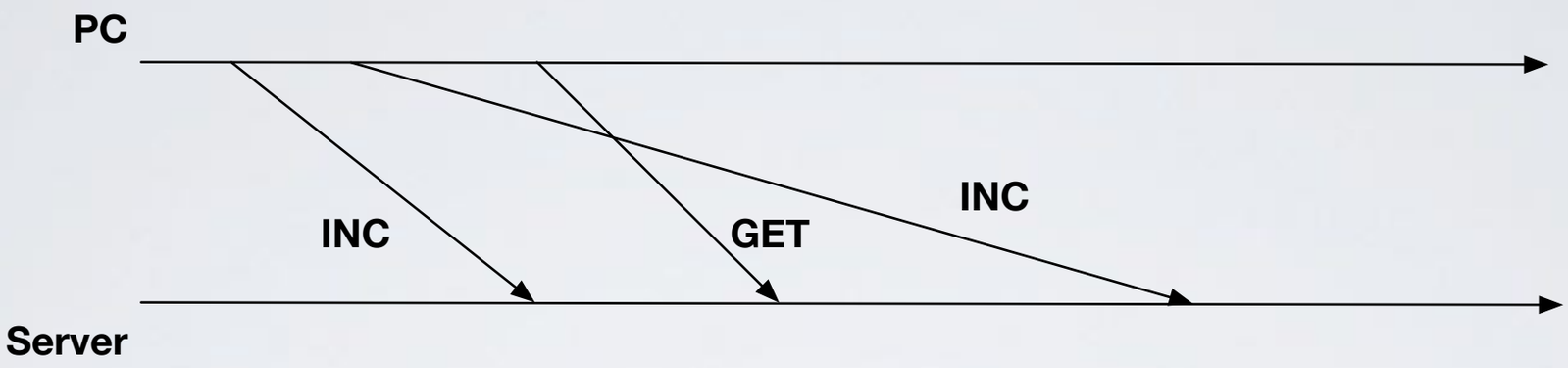
LINKS

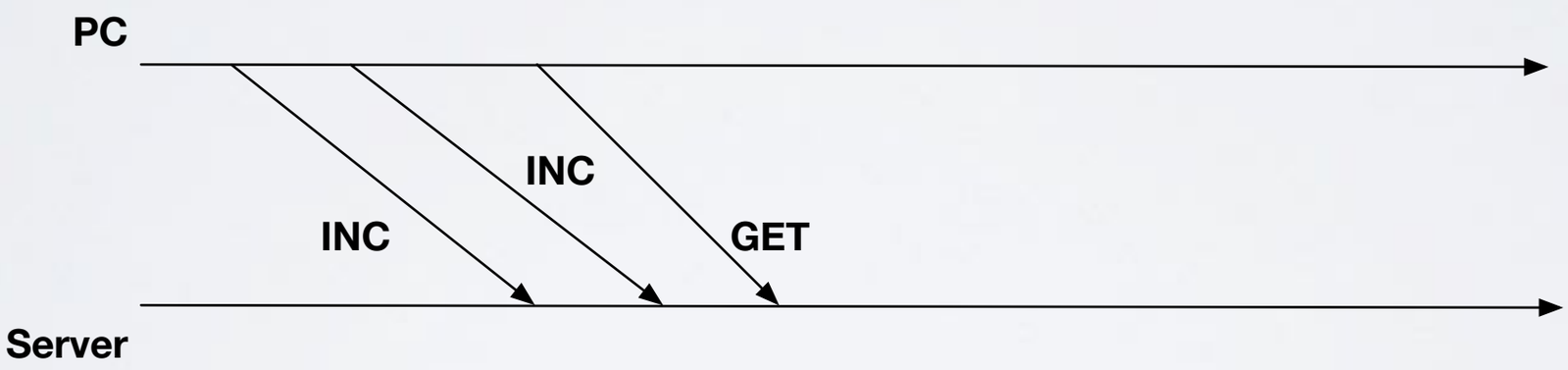


Counter 1

This scenario could happen if we get messages out of order

LINKS





But even if we get it in order, the scenario happen
If we use the code of Perfect in the MainBook

LINKS - CODE OF THE PERFECT

upon event $\langle\ pl,\ Init\ \rangle$ **do**
    $delivered := \emptyset$;

upon event $\langle\ pl,\ Send\ |\ q, m\ \rangle$ **do**
    trigger $\langle\ sl,\ Send\ |\ q, m\ \rangle$;

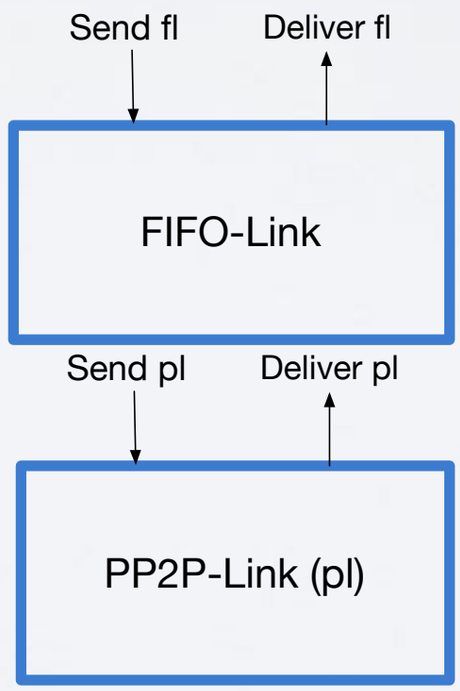
upon event $\langle\ sl,\ Deliver\ |\ p, m\ \rangle$ **do**
    if $m \notin delivered$ **then**
        $delivered := delivered \cup \{m\}$;
        trigger $\langle\ pl,\ Deliver\ |\ p, m\ \rangle$;

Messages have to be unique

If we get the same message two times we deliver
It just once.

FIFO - LINK

- Properties:
  - PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.
  - PL2: (No duplication) A message is delivered at most once.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.
  - **(FIFO)** if a process p delivers m before m', then m was sent before m'.



A FIX FOR BOTH PROBLEMS

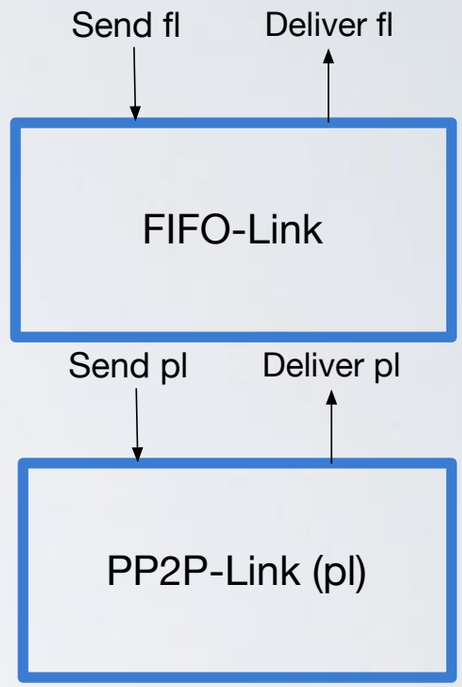
Algorithm 5 FIFO Links - client p_i

1: **upon event** INIT
2: $Pending = \emptyset$
3: $seq = 0$ ▷ Send sequence number
4: $del_seq = 0$ ▷ Rcv sequence number

5: ▷
6: **upon event** SEND FL($\langle p_d, m \rangle$)
7: SEND PERFECT($p_d, < m, seq >$)
8: $seq = seq + 1$

9: **upon event** DELIVERY FROM PERFECT($\langle p_s, < m, seq > \rangle$)
10: $Pending = Pending \cup \{\langle p_s, < m, seq > \rangle\}$

11: **upon event** $\exists \langle p_s, < m, send_seq > \rangle \in Pending$ SUCH THAT $send_seq = del_seq$
12: DELIVERY FL(p_s, m)
13: $Pending = Pending \setminus \{\langle p_s, < m, send_seq > \rangle\}$
14: $del_seq = del_seq = +1$ **typo: here you have to increment del_seq**



- **(FIFO)** if a process p delivers m before m', then m was sent before m'.

FORMAL PROOF

- Properties:
  - PL1: (Reliable delivery) If a correct process p sends m to q, then q eventually delivers m.
  - PL2: (No duplication) A message is delivered at most once.
  - FL3: (No creation) If some process q delivers a message m with sender p, then m was sent by p to q.
  - The three above holds directly from the use of the perfect channel.

  - **(FIFO)** if a process p delivers m before m', then m was sent before m'.

FORMAL PROOF

- **(FIFO)** if a process p delivers m before m', then m was sent before m'.

- Proof: By contradiction, suppose p delivers in reverse order m' and then m, even if m was sent before m'.
- This means that timestamp(m) < timestamp(m').
- Let t be the instant at which p delivers m', this implies that del_seq=timestamp(m').

```
11: upon event ∃⟨p_s, < m, send_seq >⟩ ∈ Pending SUCH THAT send_seq = del_seq
12:     DELIVERY FL(p_s, m)
13:     Pending = Pending \ {⟨p_s, < m, send_seq >⟩}
14:     del_seq = del_seq = +1   typo: here you have to increment del\_seq
```

- This is a contradiction, because for del_seq to be equal to timestamp(m'), we should have had del_seq=timestamp(m) (remember that del_seq grows by one unit each delivery). Thus m has been delivered before t.